Technical University of Denmark
Department of Applied Mathematics and Computer Science

02170 Database Systems

Written Examination 2023

Exam date:	May 2023
Duration:	3 hours
Aids:	All written materials permitted. No internet.

Note: The exam consists of two parts. Part 1 is SQL (written queries). Part 2 is multiple choice theory. The parts can be solved in any order.

Scenario: A Database for an Airline

A database named `Airline` is considered. The database contains information about flights offered by an airline company named *Noob Airlines*, passengers on the flights, and customers. Below you can find the relation schemas for the five relations in the database, and a relation instance of each of them.

Airport(airportID, city) **Person**(personID, name, speed)

Plane(planeID, capacity)

Flight(flightID, dayTime, duration, fromAirport, toAirport, planeID, departureDate, latestDepTime)

FlightPassengers(flightID, customerID, price)

Airport: Each airport has a unique *airportID* and the city in which the airport is situated.
Plane: Each plane has a unique *planeID* and a capacity stating the number of passengers that can be seated in the plane.
Flight: Each flight has a unique *flightID*, a departure date (*dayTime*) and a departure time (*latestDepTime*) (in the CET time zone), and a duration. It also has a departure airport (*fromAirport*) and a destination airport (*toAirport*). A specific plane (*planeID*) can be allocated to the flight. Initially no plane has been allocated, therefore *planeID* can be NULL.
Customer: Before a customer can be booked on a flight, the customer must be registered with a unique *customerID* and a name.
FlightPassengers: In this table it is registered which customers are booked as passengers on which flights, and which individual price each passenger of a flight has paid for that flight.

Airport

airportID	city
CPH	Copenhagen
MXP	Milano
ARN	Stockholm
OSL	Oslo
LHR	London

Plane

planeID	capacity
P1	180
P2	220
P3	150

Customer

customerID	name
C1	Alice
C2	Bob
C3	Carol
C4	Dave

Flight

flightID	dayTime	duration	fromAirport	toAirport	planeID
SK101	2023-05-01 08:00	120	CPH	MXP	P1
SK102	2023-05-01 10:00	90	MXP	CPH	P1
SK103	2023-05-02 09:00	110	CPH	ARN	P2
SK104	2023-05-02 14:00	80	ARN	CPH	NULL
SK105	2023-05-03 07:00	130	CPH	LHR	P3
SK106	2023-05-03 15:00	75	OSL	CPH	P2

FlightPassengers

flightID	customerID	price
SK101	C1	1200
SK101	C2	1500
SK101	C3	900
SK102	C1	1100
SK103	C2	800
SK103	C4	1300
SK105	C3	1250
SK105	C4	1250

Part 1 — SQL Queries

Write at most one SQL query per question. Place your answer in the answer file.

Question 1

Write an SQL query that returns a table containing the `flightID` of those flights that are flying to **Milano** city.

Question 2

Write an SQL query that returns a table containing the `flightID` of flights for which the highest price (paid by a passenger) is **less than 1300**.

Question 3

Write an SQL query that returns a table containing the `customerID` and the **number of free seats** for each flight which goes **from the CPH airport** and which has got a plane allocated.

Hint: It is assumed that each passenger booked on a flight will be assigned exactly one seat on that flight. Furthermore, the airline sometimes overbooks flights; in such a case the number of free seats will be negative.

Question 4

Define an SQL trigger named `Flight_Before_Insert`, which automatically raises a signal when a new row having the **same `fromAirport` and `toAirport`** is attempted to be inserted in the `Flight` table.

Question 5

Show an example of an SQL statement which **causes this trigger to raise a signal**.

Part 2 — Multiple Choice Theory

Select the single best answer for each question unless stated otherwise.

Question 1: ER Modelling

Consider the following relation schemas:

- **Airport**(airportID, city)
- **Flight**(flightID, fromAirport, toAirport, planeID)

Which of the following ER diagrams correctly depicts the relationship between `Airport` and `Flight`?

(Multiple ER diagram options — see original exam)

Question 2: ER Design

The database is extended to also track which customers have a *FrequentFlyerCard*. Consider the partial ER model below.

Question 2.1: Which of the following statements is true / false?

a) Every customer has a FrequentFlyerCard.

b) There are FrequentFlyerCards that are not assigned to any customer.

c) A customer can have multiple FrequentFlyerCards.

d) Every FrequentFlyerCard belongs to exactly one customer.

e) There are customers that have been assigned to multiple cards.

Question 2.2: Transformation ER to Relational Model
Which of the following options is the correct transformation to the relational model?

a) *[option a — see original exam]*

b) *[option b — see original exam]*

c) *[option c — see original exam]*

d) *[option d — see original exam]*

Question 3: Normalization

Consider a relation with schema $R(A, B, C, D, E)$ where the primary key is not shown.
The following functional dependencies are assumed:

$$FD = \{A \rightarrow BCD,\ CD \rightarrow E\}$$

Question 3.1: Primary Key

What is/are the candidate key(s) of R?

Question 3.2: Normal Form

What is the highest normal form of R?

Question 3.3: Normalization

Normalize R into third normal form (3NF), if it is not already in 3NF.

Question 4: Integrity Constraints

Question 4.1: Data Integrity

Name two options to enforce that a user can never insert a new row into `Flight` where `fromAirport = toAirport`.

Question 4.2: Foreign Key Actions

Which referential action would you pick to ensure that a flight cannot be deleted from the database while it still has passengers booked on it (i.e., there is a row in `FlightPassengers` that references it)?

Question 5: Indexing and Hashing

Assume that there is exactly one index on the `Flight` table (no automatic indexes were created), created via:

```
CREATE INDEX flight_idx ON Flight(fromAirport) USING HASH;
```

Which of the following SQL queries is sped up by this index?

a) `SELECT * FROM Flight WHERE fromAirport <= 'CPH';`

b) `SELECT * FROM Flight WHERE fromAirport = 'CPH' AND toAirport = 'MXP';`

c) `SELECT * FROM Flight WHERE fromAirport > 'CPH' AND toAirport = 'MXP';`

d) `SELECT * FROM Flight WHERE toAirport = 'MXP';`

e) `SELECT * FROM Flight WHERE flightID = 'SK101';`

Question 6: Formal Query Languages

Which of the following options in domain calculus is the equivalent of the relational algebra statement below?

$$\Pi_{flightID}\,(\sigma_{city='Milano'}\,(Flight \bowtie Airport))$$

a) $\{flightID \mid \langle flightID, fromAirport, toAirport\rangle \in Flight \land \langle toAirport, \text{'Milano'}\rangle \in Airport\}$

b) $\{flightID \mid \exists fromAirport, toAirport\,(\langle flightID, fromAirport, toAirport\rangle \in Flight \land \langle toAirport, \text{'Milano'}\rangle \in Airport)\}$

c) $\{flightID \mid \exists fromAirport, toAirport\,(\langle flightID, fromAirport, toAirport\rangle \in Flight) \land \exists toAirport\ (\langle toAirport, \text{'Milano'}\rangle \in Airport)\}$

d) $\{flightID \mid \exists fromAirport, toAirport, city\ (\langle flightID, fromAirport, toAirport\rangle \in Flight\ \land\ \langle toAirport, city\rangle \in Airport)\}$

— End of Exam —